The Network Layer

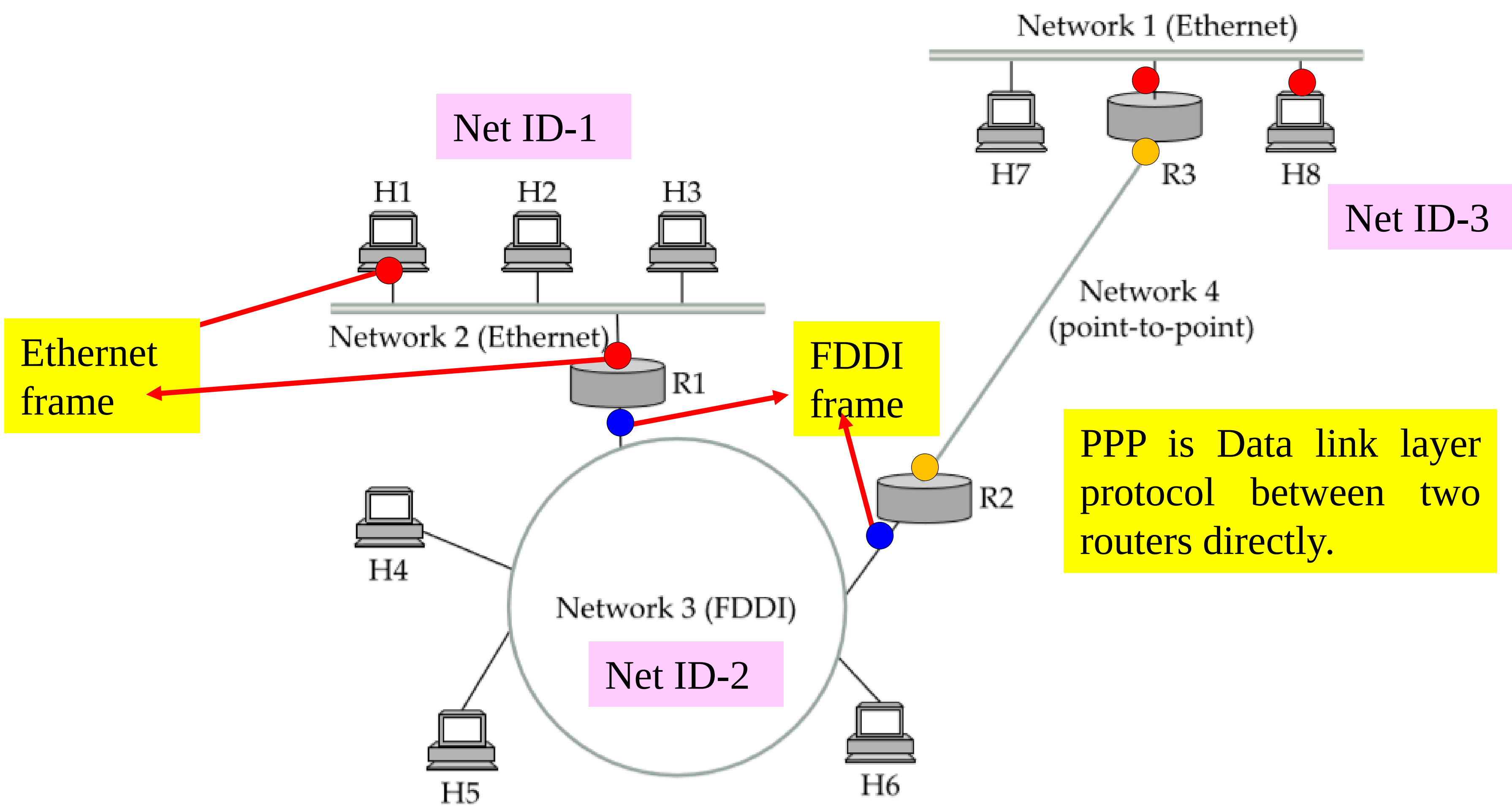
- ✓ The **network layer** is responsible for the source-to-destination delivery of a packet (called IP packet), possibly across multiple networks (homogeneous or heterogeneous) networks called *internetworking*.

- ✓ In case of homogeneous networks (each network use same protocol) packet is passed through router and that of through heterogeneous networks (dissimilar protocol is used by network, for example 4G LTE (WAN) network to IPv6 network) the job is done through gateway router.

IoT Gateway: Connects WSN to an IP-based internet network.
VoIP Gateway: Converts telephony voice signal into IP packets for the internet

- ✓ Sometimes the network is heterogeneous at datalink layer but becomes homogeneous at network layer. Ethernet uses CSMA/CD but FDDI uses token ring protocol. They are different at datalink layer or in context of frame but same at network layer in context of IP packet.

In this figure, we see three types of networks: Ethernets, an FDDI (Fibre Distributed Data Interface) ring, and a point-to-point link. The nodes that interconnect the networks are called *routers*.



A simple internetwork: H*n* = host; R*n* = router

Router(config)# interface fddi 4/0

Router(config)# interface fa 4/0

Figure below shows how hosts H1 and H8 are logically connected by the internet in including the protocol graph running on each node. Let H1 transmits a TCP packet to node H8.

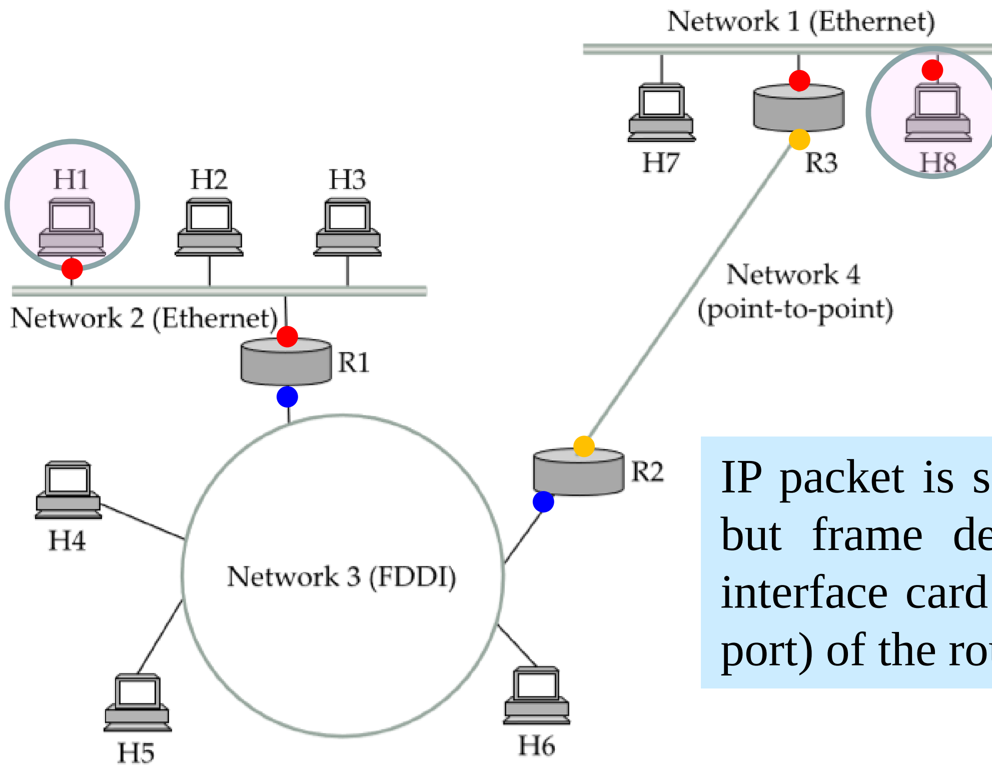


Figure (a)

IP packet is same for all the network but frame depends on the network interface card (Ethernet port or FDDI port) of the router

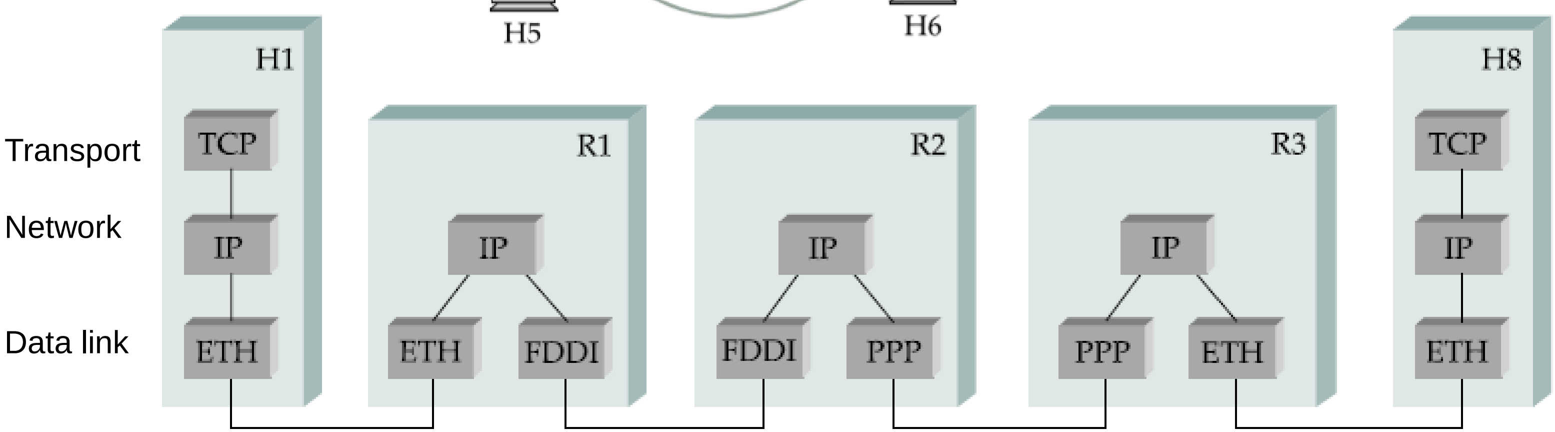


Figure (b)

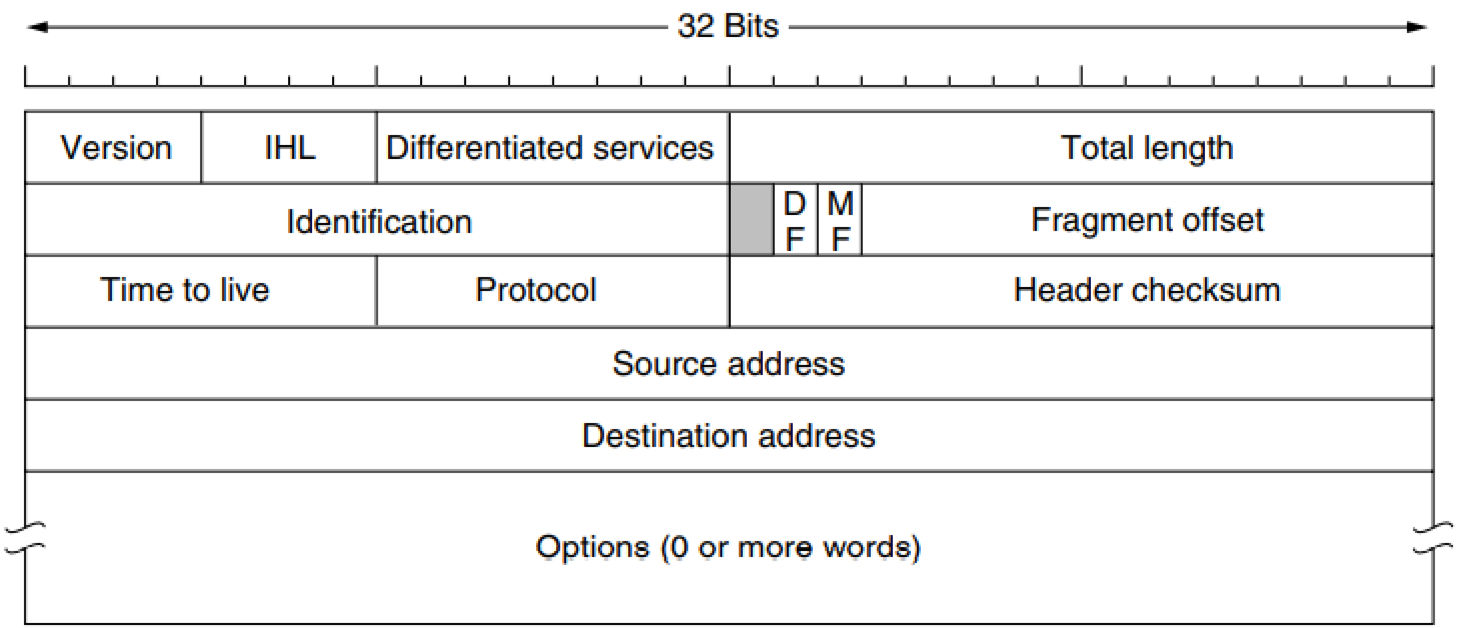
IP Packet

- ✓ We need a global addressing scheme to route packet through interconnected network; we called this logical addressing as IP address under TCP/IP protocol suite.
- ✓ In IPv4 (IP version 4) the length of the address is 32 bits and for new generation of IP or IPv6 (IP version 6) the length of address is 128 bits. This section deals with different fields of IP packet.
- ✓ The IP packet, like most packets, consists of a header followed by a number of bytes of data.

IPv4 packet:

Version number (4 bits)
Indicates the version of the IP protocol
Typically “4” (for IPv4), and sometimes “6” (for IPv6)

Internet Header Length (IHL)
The header length is not constant, IHL field indicates the number of 32 bit words in the header



IP Packet Structure (IPv4)

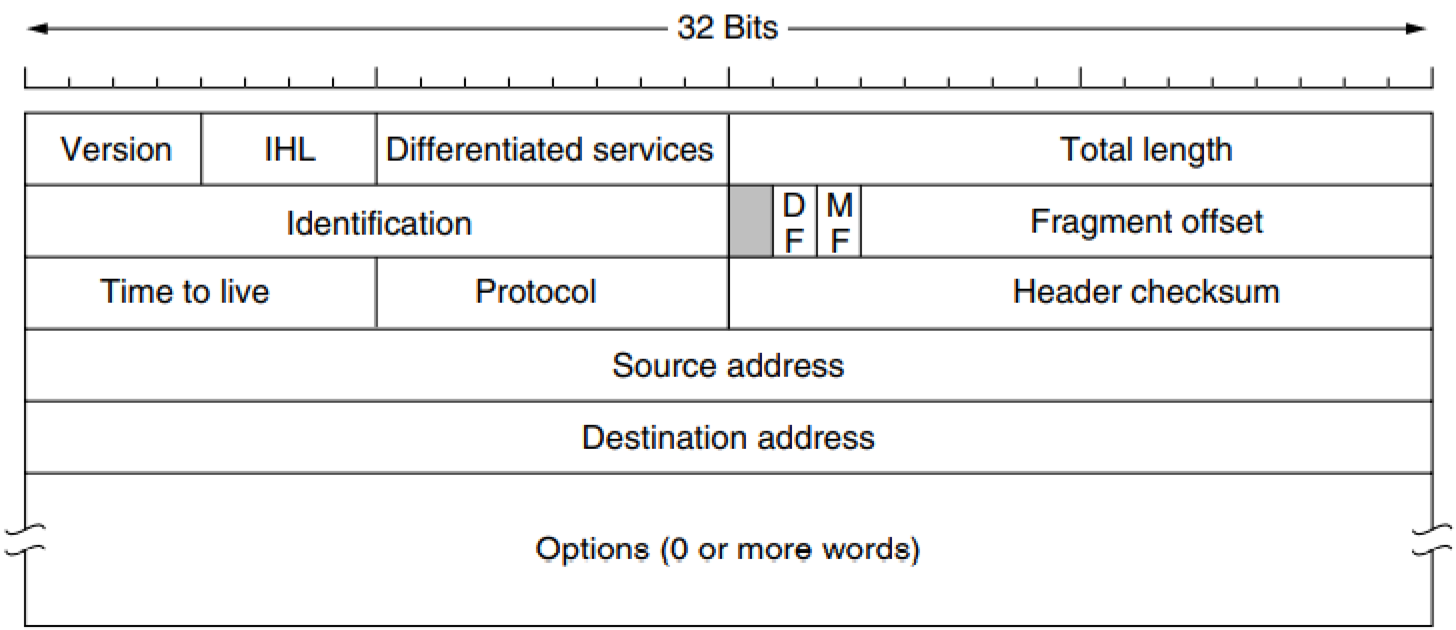
Differentiated Services (8 bits)

❖It is intended to distinguish between different classes of service. Various combinations of reliability and speed are possible.

❖For digitized voice, fast delivery is more essential than accurate delivery.

❖For file transfer, error-free transmission is more important than fast transmission.

❖This field is related to three parameters: {Delay, Throughput, Reliability} in communication.

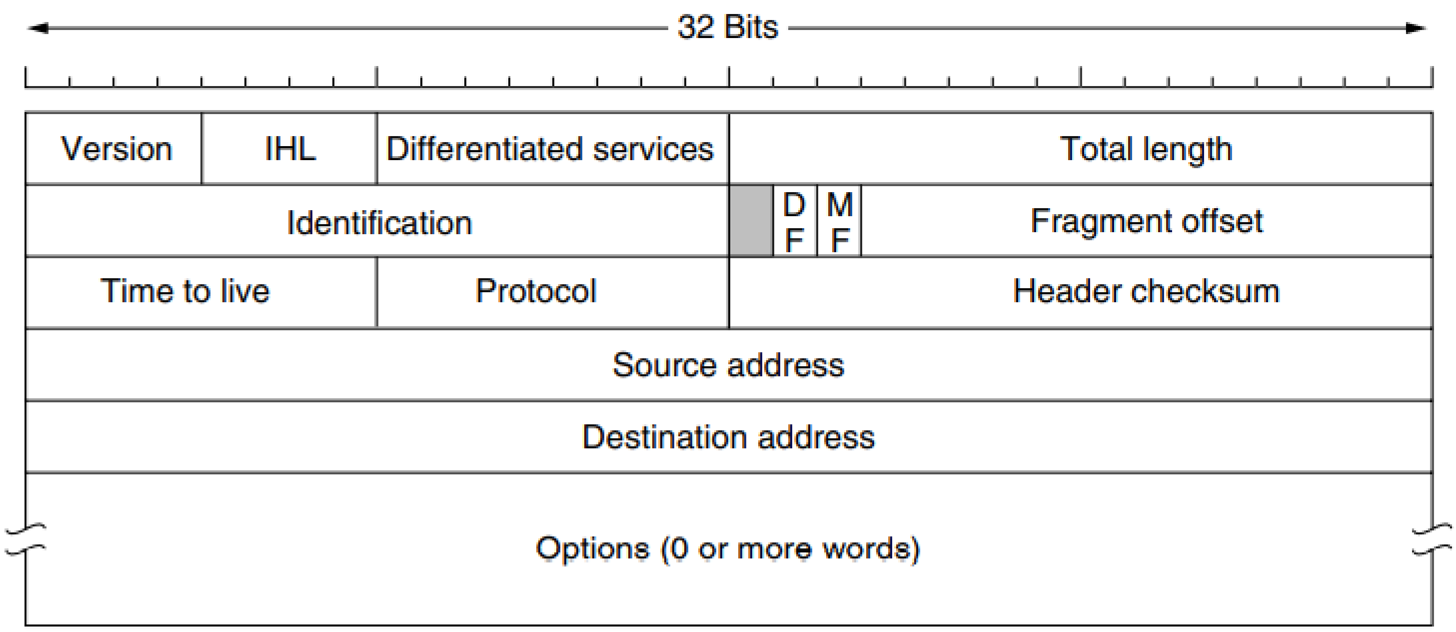


Total length (16 bits)

- Number of bytes in the packet
- Maximum size is 65,535 bytes (2^{16} -1)

Identification

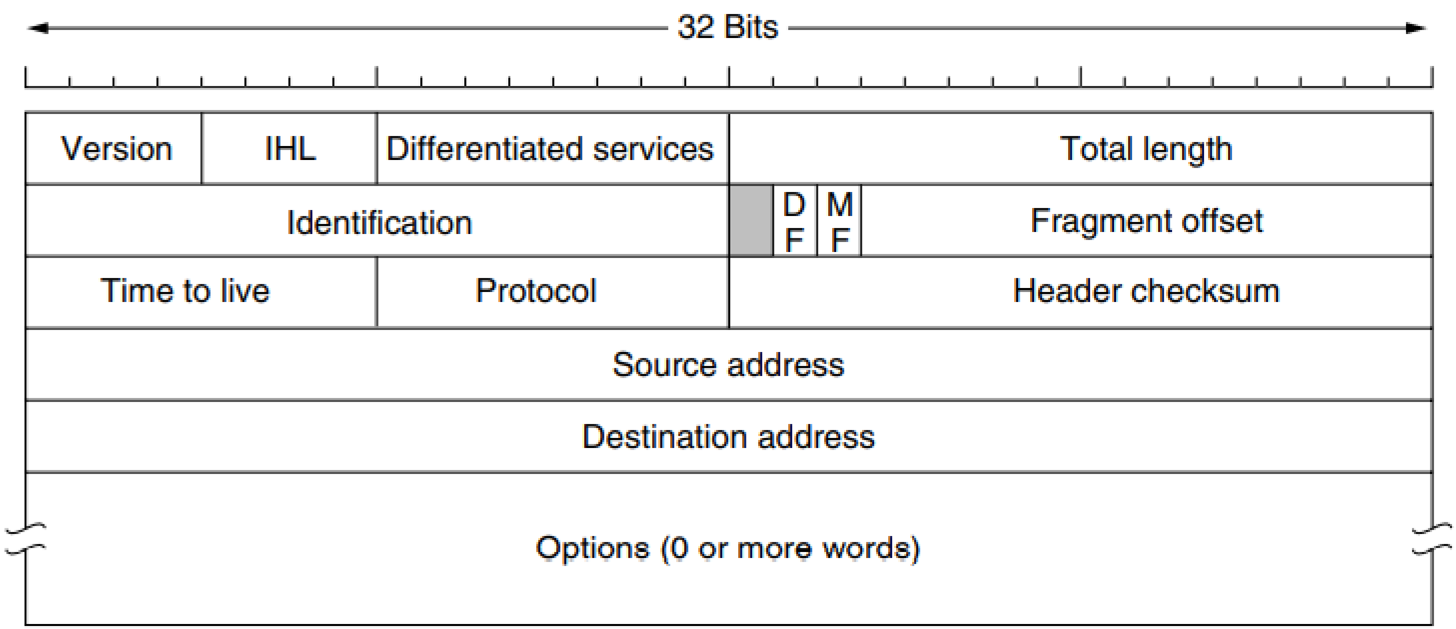
The Identification field is needed to allow the destination host to determine which datagram a newly arrived fragment belongs to. All the fragments of a datagram contain the same Identification value.



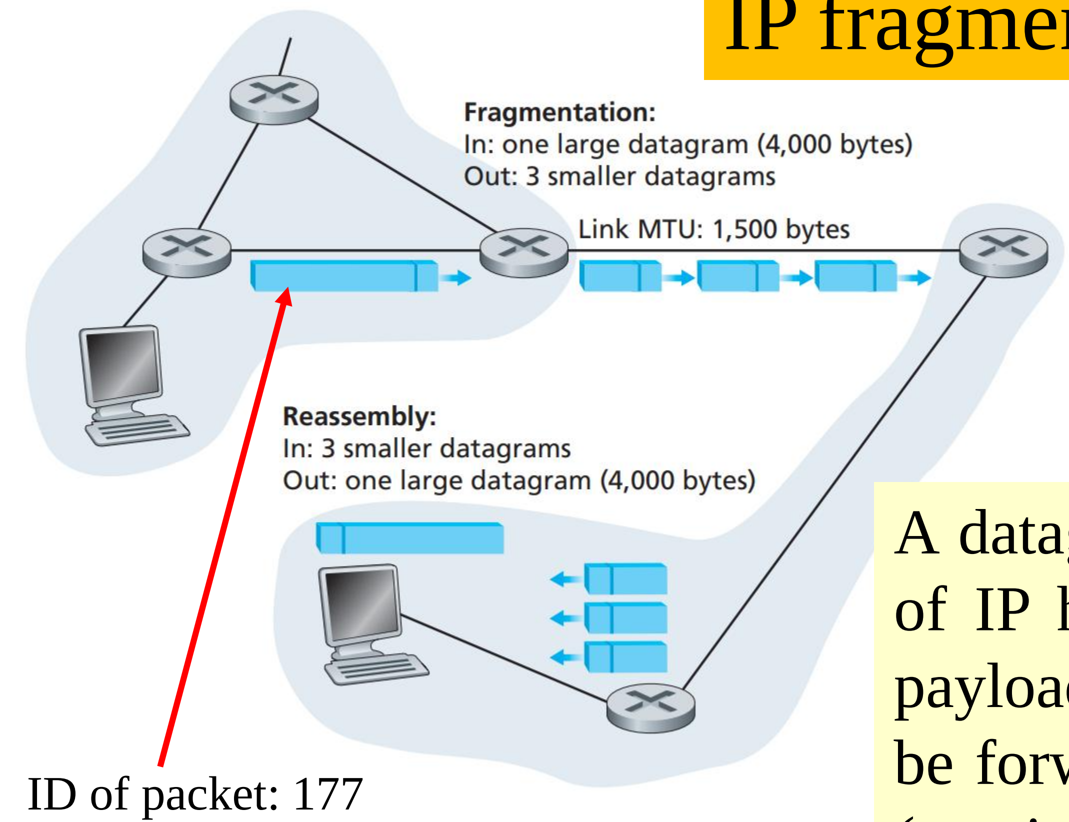
Flags : A three-bit field follows and is used to control or identify fragments. They are (in order, from high order to low order):

bit 0: Reserved; must be zero.
bit 1: Don't Fragment (DF)
bit 2: More Fragments (MF)

Fragment Offset The Fragment offset (distance) tells where in the current packet this fragment belongs i.e. its starting position. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit. Since 13 bits are provided, there is a maximum of $2^{13} = 8192$ fragments per datagram.



IP fragmentation and reassembly

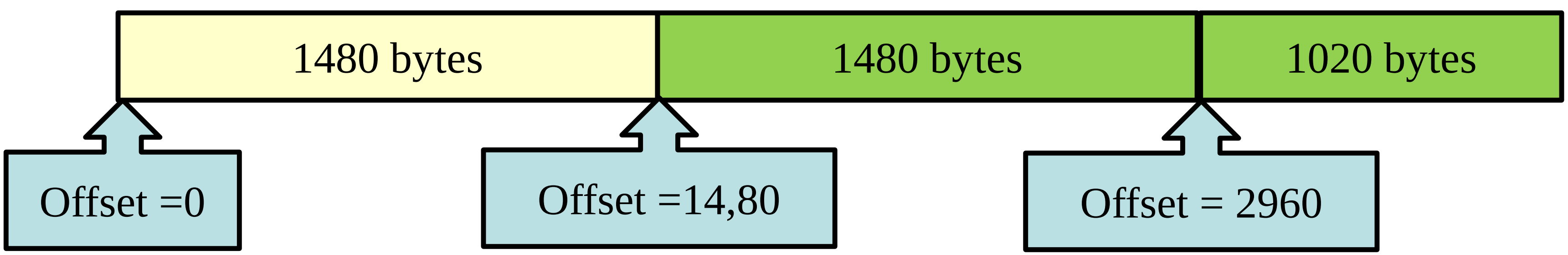


A datagram of 4,000 bytes (20 bytes of IP header plus 3,980 bytes of IP payload) arrives at a router and must be forwarded to a link with an MTU (maximum transmission unit) of 1,500 bytes.

This implies that the 3,980 data bytes in the original datagram must be allocated to three separate fragments (each of which is also an IP datagram). Suppose that the original datagram is stamped with an identification number of 177.

IP fragments

Fragment	**Bytes**	**ID**	**Offset**	**Flag**
1st fragment	1,480 bytes in the data field of the IP datagram	177	offset = 0 (meaning the data should be inserted beginning at byte 0	MF =1 (meaning there is more)
2^{nd} fragment	1,480 bytes of data	177	offset = 1,480 (data should be inserted beginning at byte 1,480)	MF =1 (meaning there is more)
3^{rd} fragment	1,020 bytes of data	177	offset = 2,960 (data should be inserted beginning at byte 2,960)	MF = 0 (meaning this is the last fragment)

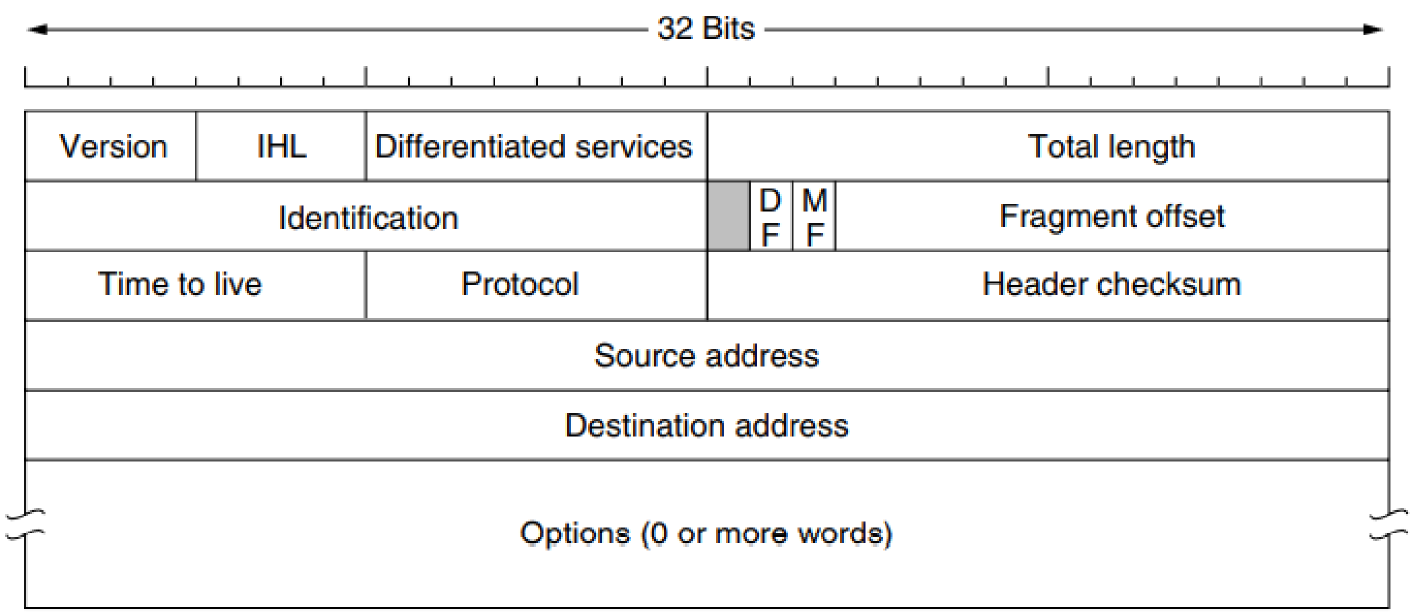


$$1480+1480 + 1029 = 3980$$

Time-To-Live (8 bits)

- ✓ The time-to-live (TTL) field is used to control the maximum number of hops (routers) visited by the datagram. When a source host sends the datagram, it stores a number in this field. This value is approximately 2 times the maximum number of routes between any two hosts.

- ✓ Each router that processes the datagram decrements this number by one. If this value, after being decremented, is zero, the router discards the datagram.



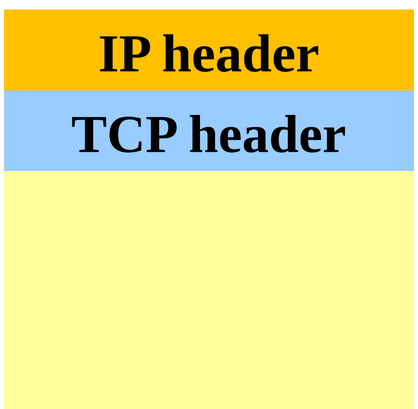
Protocol (8 bits)

The Protocol field tells it which transport process to give the packet to. TCP is one possibility, but so are UDP and some others.

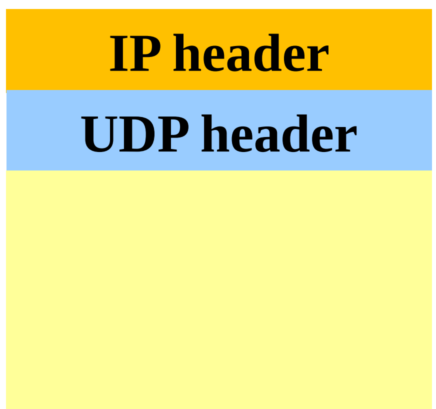
Identifies the higher-level protocol

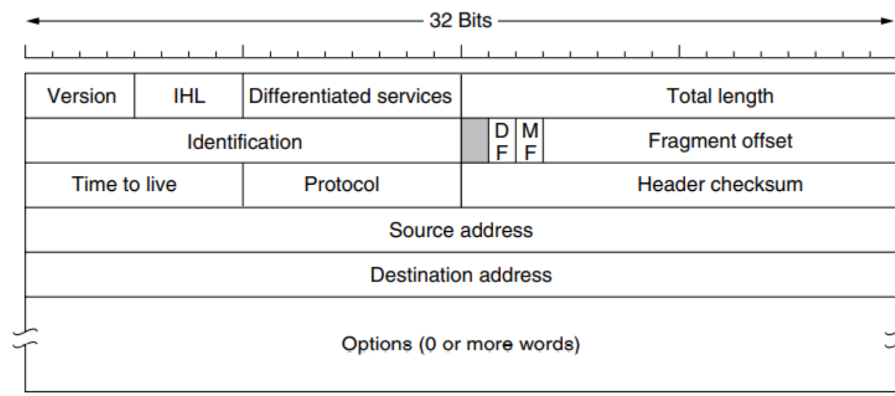
- "6" for the Transmission Control Protocol (TCP)
- "17" for the User Datagram Protocol (UDP)

Protocol = 6



Protocol = 17

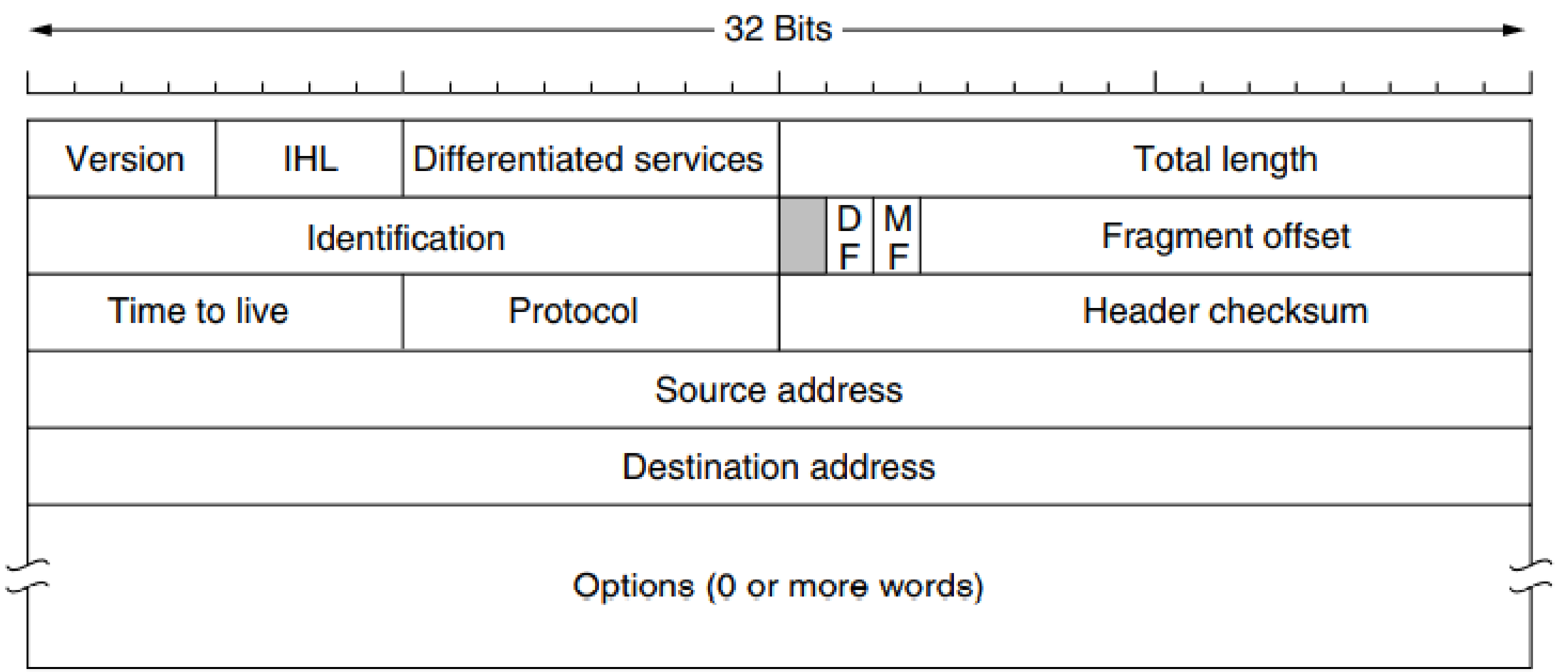




Checksum (16 bits)

- Sum of all 16-bit words in the IP packet header
- If any bits of the header are corrupted in transit

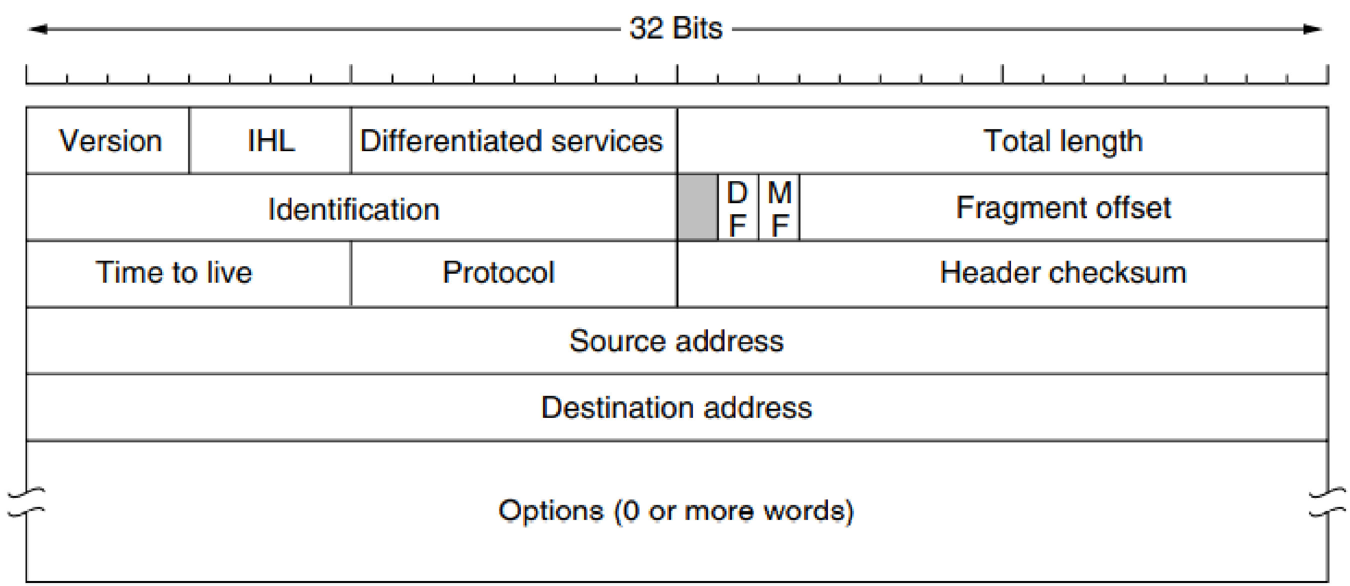
  … the checksum won't match at receiving host
- Receiving host discards corrupted packets
  - Sending host will retransmit the packet, if needed



Example: Calculating a checksum

- The header is shown in red and the checksum is underlined. **4500 0073 0000 4000 4011 b861 c0a8 0001 c0a8 00c7** 0035 e97c 005f 279f 1e4b 8180

- To calculate the checksum, we can first calculate the sum of each 16 bit value within the header, skipping only the checksum field itself. Note that the values are in hexadecimal notation. 4500 + 0073 + 0000 + 4000 + 4011 + c0a8 + 0001 + c0a8 + 00c7 = 2479C (equivalent to 149,404 in decimal).



- Convert the value 2479C to binary:

0010 0100 0111 1001 1100

The first 4 bits are the carry and will be added to the rest of the value:

0010 + 0100 0111 1001 1100 = 0100 0111 1001 1110

Next, we flip every bit (1's complement) in that value, to obtain the checksum:

0100 0111 1001 1110 becomes:

1011 1000 0110 0001

This is equal to **B861** in hexadecimal, as shown underlined in the original IP packet header.

- **Verifying a checksum**

When verifying a checksum, the same procedure is used as above, except that the original header checksum is not omitted.

4500 + 0073 + 0000 + 4000 + 4011 + **b861** + c0a8 + 0001 + c0a8 + 00c7 = 2fffd

Add the carry bits:

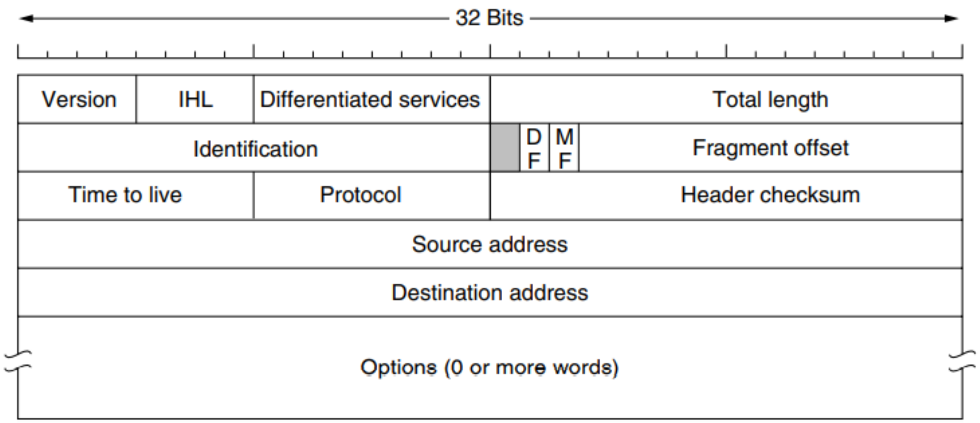
fffd + 2 = ffff

Taking the 1's complement (flipping every bit) yields 0000, which indicates that no error is detected. IP header checksum does not check for the correct order of 16 bit values within the header.

- Two IP addresses
  - Source IP address (32 bits)
  - Destination IP address (32 bits)
- Destination address
  - Unique identifier for the receiving host
  - Allows each node to make forwarding decisions
- Source address
  - Unique identifier for the sending host
  - Recipient can decide whether to accept packet
  - Enables recipient to send a reply back to source

Options

The Options field was designed to provide an opportunity to allow subsequent versions of the protocol to include information not present in the original design and to permit experimenters to try out new ideas.

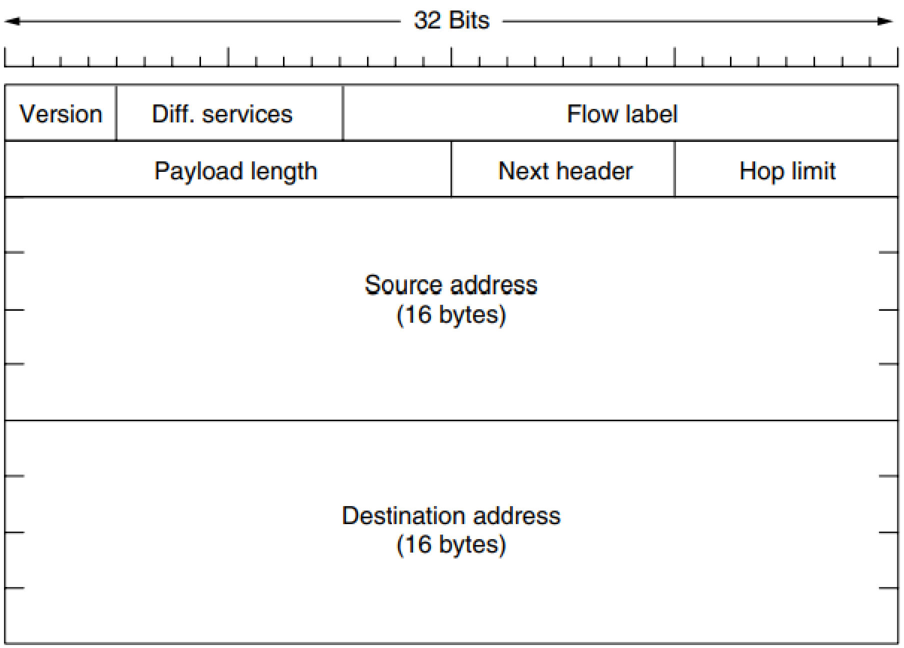


Some of the IP options

Option	Description
Security	Security Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

IPv6 Header

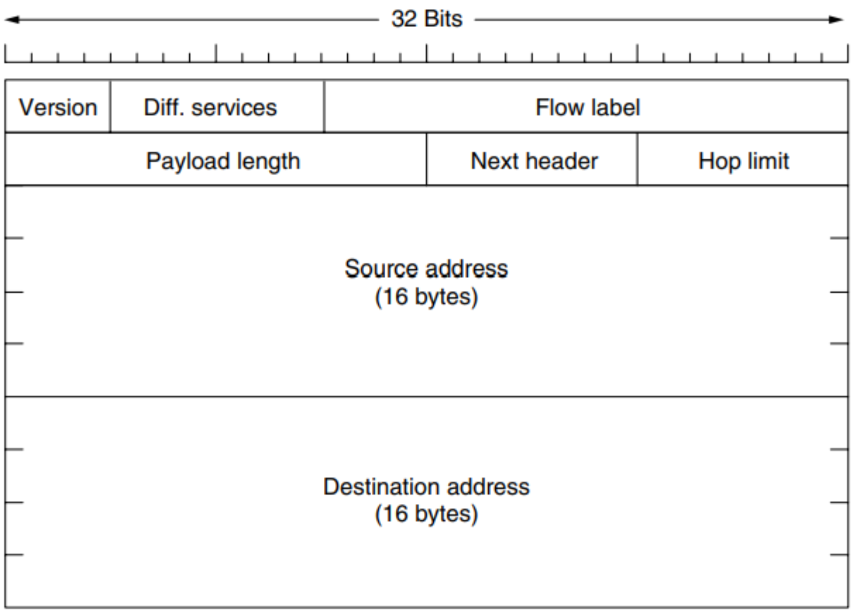
✓ The **Flow label** field provides a way for a source and destination to mark groups of packets that have the same requirements and should be treated in the same way by the network.



✓ For example, a stream of packets from one process (on a certain source host) to a process on a specific destination host might have stringent (strict condition) delay requirements and thus need reserved bandwidth.

✓ When a packet with a nonzero Flow label shows up, all the routers can look it up in internal tables to see what kind of special treatment it requires.

The **Payload length** field tells how many bytes follow the 40-byte header i.e. it measures the length of message field of Fig. below.

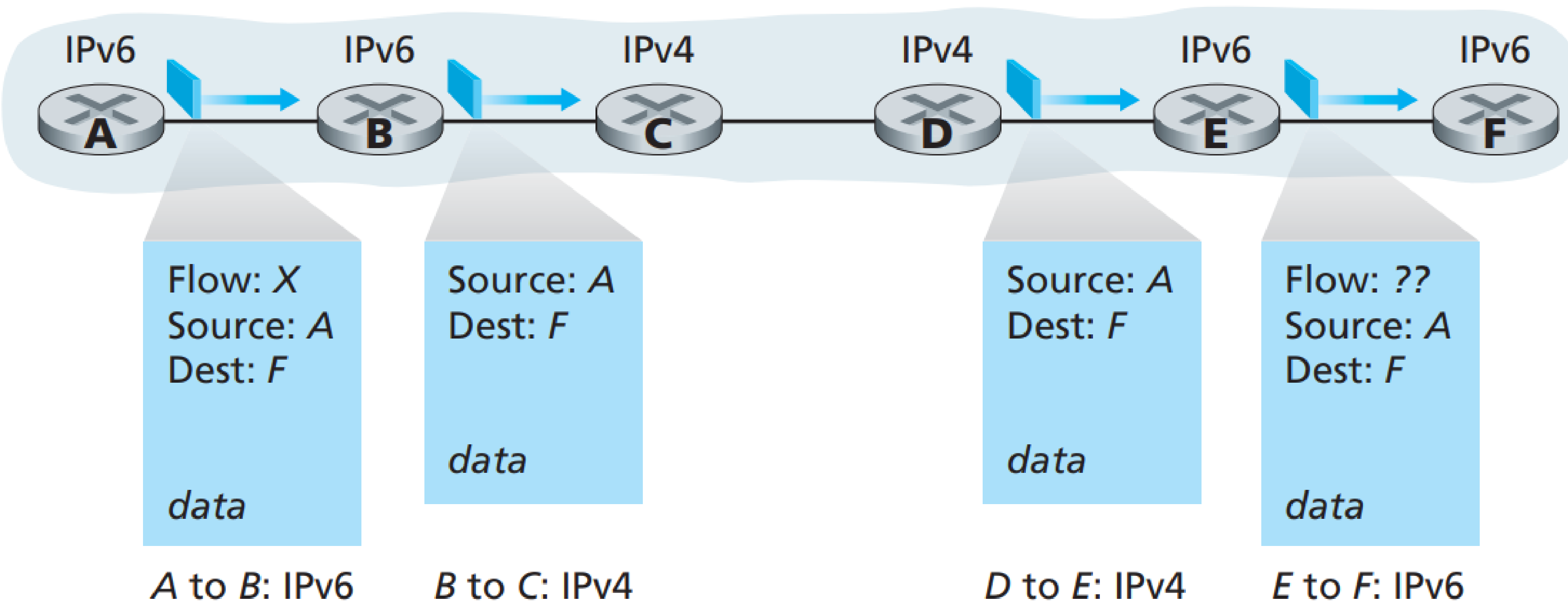


Next header field tells which transport protocol (e.g., TCP, UDP) is passed to the packet.

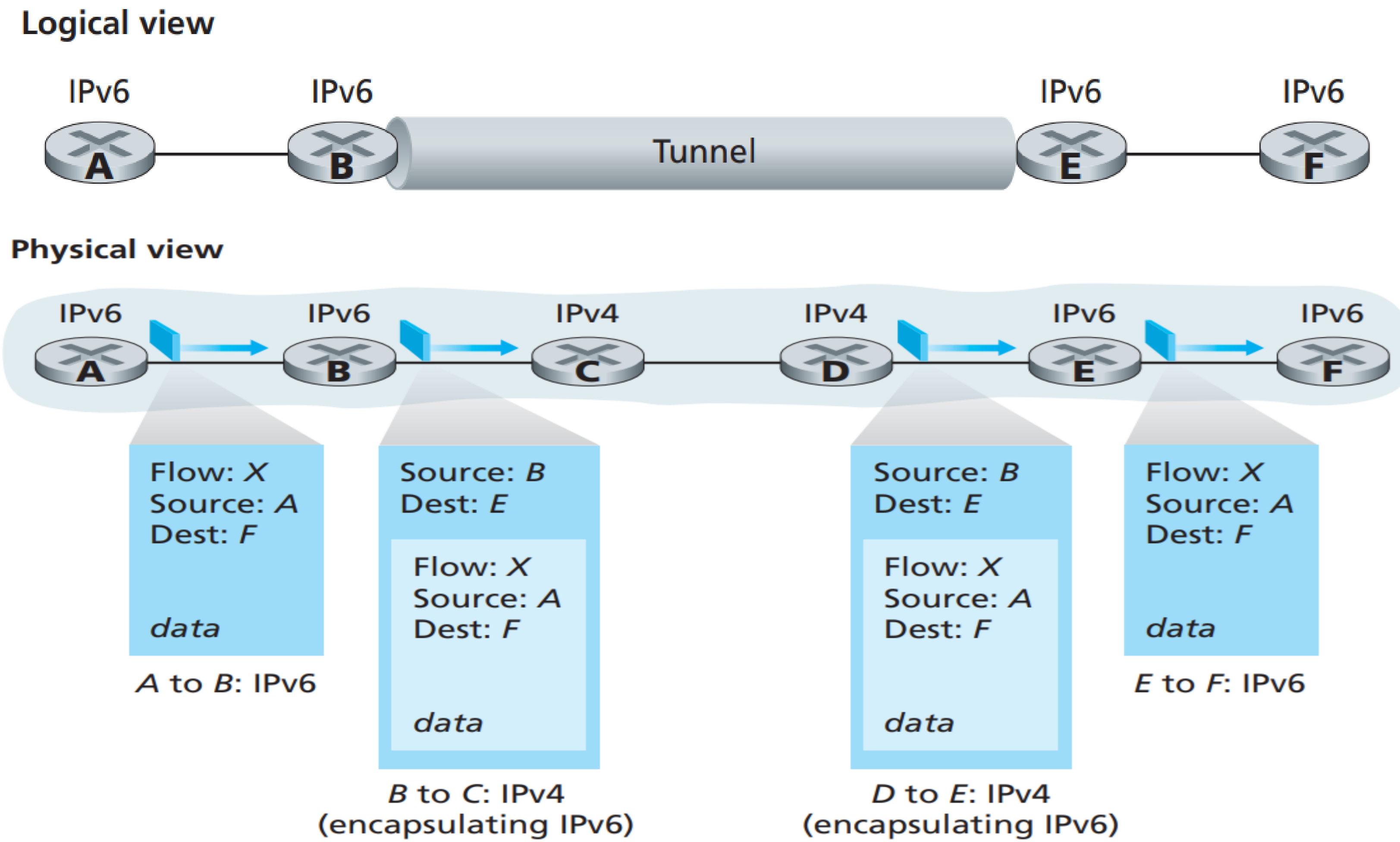
The **Hop limit** field is used to keep packets from living forever. It is, in practice, the same as the Time to live field in IPv4, namely, a field that is decremented on each hop.

IPv6 has 40-byte fixed header. The fixed-length header allows for faster processing of the IP datagram.

Transitioning from IPv4 to IPv6: Suppose Node A is IPv6-capable and wants to send an IP datagram to Node F, which is also IPv6-capable. Nodes A and B can exchange an IPv6 datagram. However, Node B must create an IPv4 datagram to send to C. The conversion from IPv6 to IPv4 is not possible due to absence of some counterpart of IPv6 in IPv4.



- ✓ Tunneling can solve the problem, allowing, for example, E to receive the IPv6 datagram originated by A.
- ✓ In this case, IPv6 node on the sending side of the tunnel (for example, B) takes the entire IPv6 datagram and puts it in the data (payload) field of an IPv4 datagram. This IPv4 datagram is then addressed to the IPv6 node on the receiving side of the tunnel (for example, E) and sent to the first node in the tunnel (for example, C).

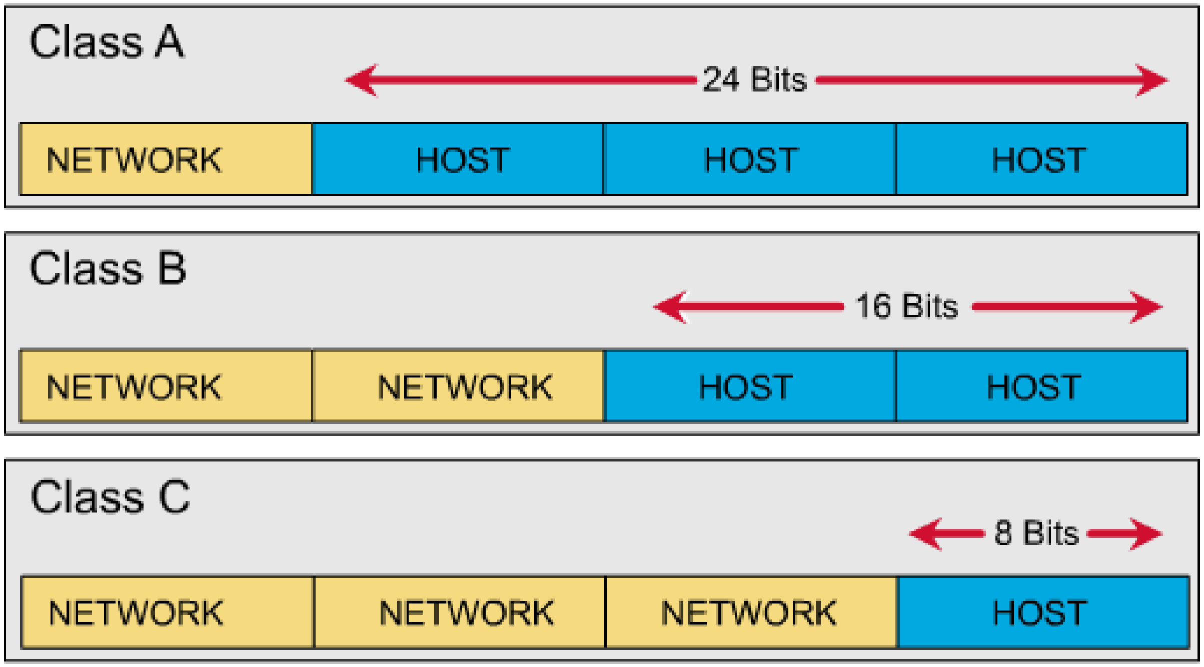


IP Addresses

✓An IP address has two components, the **network address** (Net ID) and the **host address** (Host ID).

✓A subnet mask separates the IP address into the network and host addresses (**<network> <host>**).

IPv4 address composed of 4 segments each containing 8 bits therefore entire address has 32 bits.



Class A (24 bits for hosts) $2^{24} - 2^{*}$ = 16,777,214 maximum hosts

Class B (16 bits for hosts) $2^{16} - 2^{*}$ = 65,534 maximum hosts

Class C (8 bits for hosts) $2^{8} - 2^{*}$ = 254 maximum hosts

IP address is written in dotted decimal format like: **192.168.2.21**

Subnet Mask

✓A Subnet mask is a 32-bit number that masks an IP address and divides the IP address into **network address** and **host address**. Subnet Mask is made by setting network bits to all ‘1’s and setting host bits to all ‘0’s.

IP (class C): 192.168.003.6

Subnet mask: 255.255.255.0

11111111. 11111111. 11111111. 00000000

✓A subnet mask is used to identify network address of an IP address by performing bitwise AND operation.

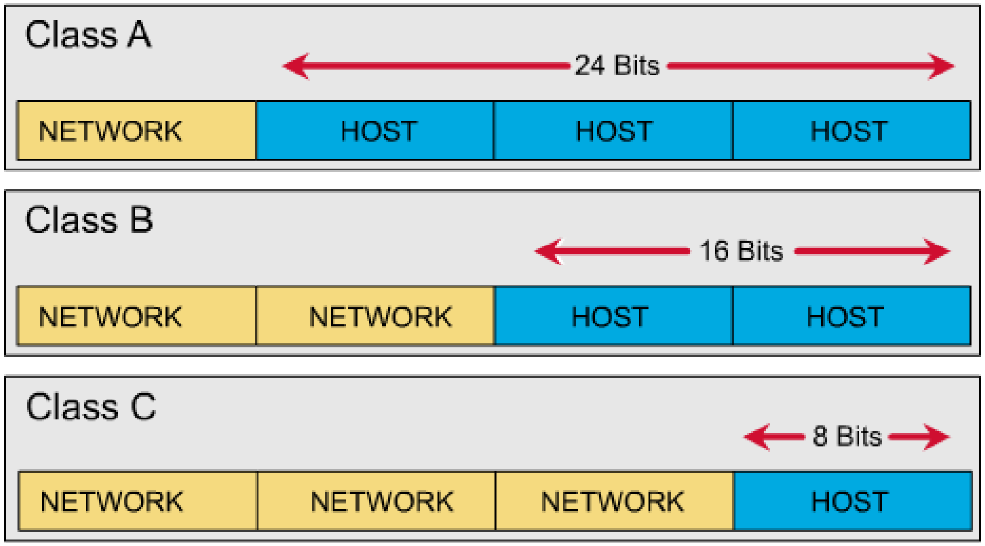
✓Within a given network, **two host addresses** are reserved for special purpose. The "0" address is assigned a **network address** and "255" or all 1, is assigned to a **broadcast address**, and they cannot be assigned to a host.

The **default** subnet mask used for class A, B, and C are:

Class A: 11111111.00000000. 00000000. 00000000
255.0.0.0

Class B: 11111111. 11111111. 00000000. 00000000
255.255.0.0

Class C: 11111111. 11111111. 11111111. 00000000
255.255.255.0



For example, we have a class C IP address of a host: **192.168.23.4** ANDing the IP address with default subnet mask **255.255.255.0** give the result of 192.168.23.0 which is the net ID part of the IP address.

11000000. 10101000. 00010111. 00000100
And
11111111. 11111111. 11111111.   00000000

11000000. 10101000. 00010111. 0000000
= 192.168.23.0

IP address AND Subnet mask = Network ID

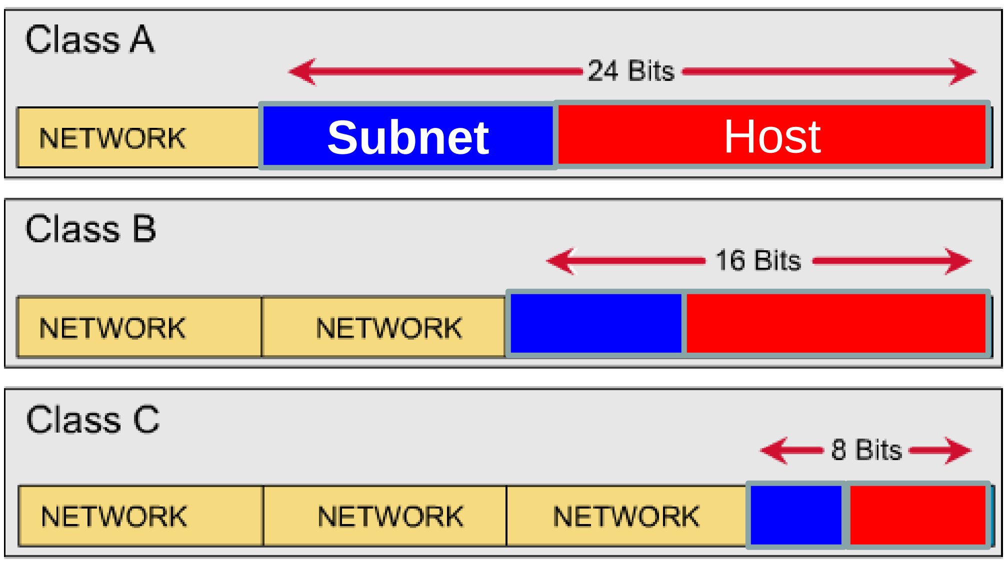
How many host is possible against the class C network ID: 192.168.23.0?

The possible hosts will be:
192.168.23.1
192.168.23.2
192.168.23.3
192.168.23.4
……………..
192.168.23.254

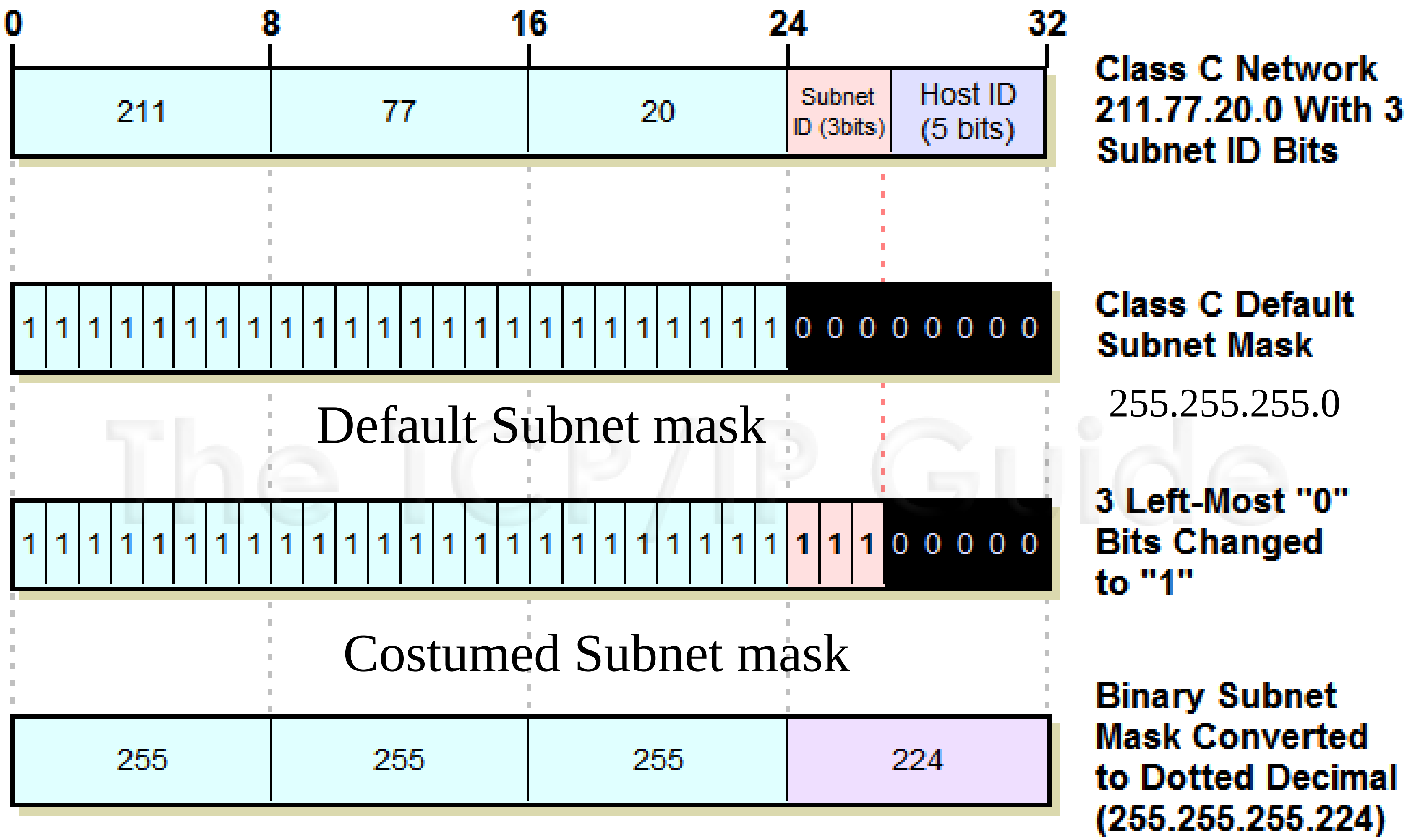
192.168.23.0 and 192.168.23.255 are excluded as mentioned before. Therefore, the number of hosts is 2^8-2 = 254

Subnet on IP address

Subnetting can further divides the host part of an IP address into a subnet and host address (**<network> <subnet> <host>**).



✓Subnetting further divides the host part of an IP address into a subnet and host address (<network> <subnet> <host>).



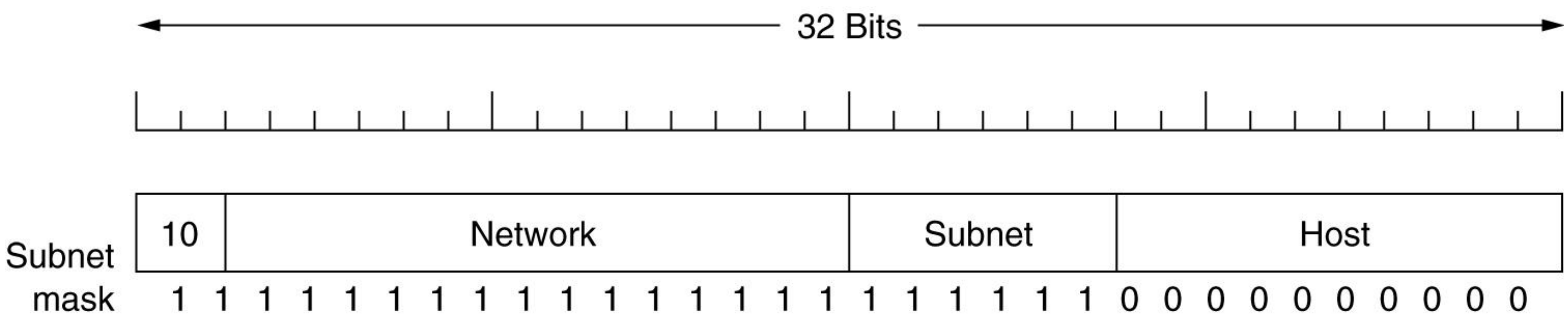
10000000 → 128
11000000 → 192
11100000 → 224
11110000 → 240
11111000 → 248

Subnet masks:

255.255.255.128
255.255.255.192
255.255.255.224
255.255.255.240
255.255.255.248

Example-1

The example of a customized subnet mask for class B is like:



A class B network subnetted into 64 subnets.

Example: Determine the number of subnet and host per subnet

Ans. The number of subnet = $2^6 = 64$

The number of hosts/subnet = $2^{10}-2= 1022$

Example-2

If the subnet mask 255.255.240.0 is used for a class B IP address then find the number of subnets and number of hosts/subnet.

Binary: 11111111 . 11111111 . 11110000 . 00000000

Decimal: 255.255.240.0

The number of hosts/subnet = $2^{12} - 2 = 4094$

The number of subnets = $2^4 = 16$

Example-2

If the subnet mask 255.255.255.192 is used for a class C IP address then fined the number of subnets and number of hosts/subnet.

The number of hosts/subnet = $2^6 - 2 = 62$

The number of subnets = $2^2 = 4$

Example-3

A class C IP address is 192.168.14.163 and the corresponding subnet mask is 255.255.255.128 Determine the maximum number of hosts per subnet and the number of subnets.

Ans. The subnet mask in both binary and decimal is like:

11111111. 11111111. 11111111. 10000000

255.255.255.128

Here 1 bits are for subnets and 7 bits for hosts.

Therefore, the number of hosts per subnet 2^7-2 = 126

The number of subnets are 2^1= 2

IPv6

IPv6 uses 128 bit and expressed in 32 hexadecimal numbers like:

EFAC: BA89: 7529:AFDC: 92AF:8654:1293 :29A2

After every 4 digits a colon ‘:’ is used therefore 32 digits + 7 colons = 39 characters hence called addressing system of 39 characters.

In some addresses huge number of zeros exits like:

DFAC: 0000: 0000:0000: 0009:03AC1:5923 :FEA2

can be expressed as:

DFAC: 0: 0:0: 9:3AC1:5923 :FEA2

or

DFAC: : 9:3AC1:5923 :FEA2

Double colon can be used only once in a IPv6 address.